Multiplicity in a CSTR Cascade

Notation: n = number of stages; OP 1 = extinguished branch, OP 2 = unstable intermediate branch, and OP 3 = ignited branch.

4. Reduction from 3^n to 4n - 1

If each stage could independently occupy any of three operating points, an n-stage cascade would have 3^n combinations. However, the stages are coupled: upstream conversion changes the composition, temperature and remaining heat-release potential entering the next stage.

Under the paper's **two-key-stage reaction-front structure**, the relevant patterns are:

Pattern family	Number
All stages on OP 1	1
One OP 2 stage, or one OP 3 stage	n + n
One adjacent OP 2-OP 2 pair, or OP 2-OP 3 pair	(n - 1) + (n - 1)

All other stage labels are 1. The two-stage pairs are fragile and exist only over a narrow operating range. A single key stage has n possible positions; an adjacent pair has n - 1.

$$\mathbf{N(n) = 1 + 2n + 2(n - 1) = 4n - 1}$$

Example: For n = 2, the retained patterns are 11, 12, 13, 21, 31, 22 and 23: seven rather than nine. The paper excludes 32 and 33 because an upstream ignited stage suppresses downstream multiplicity.

This is a maximum candidate-pattern count under the stated pathway structure, not the number of solutions guaranteed to coexist at every operating condition. Adjacency and the allowed pairs are essential; “at most two stages” alone does not yield 4n - 1.

5. Why at most two key stages are relevant

An ignited stage consumes much of the available reactant. Consequently, the downstream adiabatic temperature rise, $\Delta T_{ad,i}$, decreases and its energy-balance line becomes steeper:

$$X_i^e(T) = m_i(T - T_{op,i}), \quad m_i = (1 + St/n)/\Delta T_{ad,i}.$$

Here St is the overall Stanton number. If the maximum slope of the downstream mass-balance curve is smaller than m_i, then

$$\mathbf{d[X_i^m(T) - X_i^e(T)]/dT < 0.}$$

The difference is strictly decreasing and has at most one zero; hence the stage has at most one steady state for its fixed inlet. If this condition holds for every subsequent reachable inlet, the downstream continuation is unique.

A first stage on OP 2 can leave enough reaction potential for a second stage on OP 2 or OP 3, explaining the fragile 22 and 23 patterns. In the paper's relevant solutions, further stages have a unique continuation, localizing branching to at most two adjacent key stages.

Scope: The two-key-stage result is physically motivated and numerically supported for the investigated cascade. It is not a universal theorem for arbitrary CSTR cascades; a rigorous model-specific proof requires verifying the downstream slope condition above.

Reference: Bremer, J. & Sundmacher, K. (2021), *Novel Multiplicity and Stability Criteria for Non-Isothermal Fixed-Bed Reactors*, Frontiers in Energy Research 8:549298. Eqs. (11)-(12), pp. 10-13, Figs. 7-12. DOI: 10.3389/fenrg.2020.549298.